

WORK EXPERIENCE AND RESEARCH

Research Assistant

Apr. 2024 - Present

Nepal Applied Mathematics and Informatics Institute for research

Lalitpur, Nepal

Supervisor: Dr. Binod Bhattarai

- Working on research projects focused on **trustworthy machine learning**.
- Developed an **OOD detection method** for pathology segmentation in volumetric medical images, specifically MRI and CT scans, based on **self-supervised learning** with synthetic anomalies, leading to a **5th-rank** in the MICCAI 2024 challenge.

Undergraduate Research Intern

Aug. 2023 - Mar. 2024

Nepal Applied Mathematics and Informatics Institute for research

Lalitpur, Nepal

Supervisor: Dr. Binod Bhattarai

- Worked on an **Active Learning framework** to determine the most effective **uncertainty-based sampling technique** to address dataset imbalance in a multi-class object detection task.
- Trained deep learning models and conducted an ablation study for detecting gait abnormalities in warehouse environments.

EDUCATION

Pulchowk Campus - Institute of Engineering, Tribhuvan University

Nov. 2019- Apr. 2024

B.E in Electronics, Communication and Information

- Achieved Rank 104 in the university entrance examination out of nearly 18,000 candidates - **top 0.6%**, and received the scholarship provided by the Government of Nepal covering full tuition expenses
- Graduated with **75.73%**, ranked in the top 5% of students, and as the **top-performing female student** in the program.

PUBLICATIONS

- **Chhetri, A.**, Shrestha, P., Rana, R., Gurung G., Philip. S, Gyawali, P., & Bhattarai, B. (2026). **When Confidence Lacks Concepts: Interpretable OOD Detection via Representation Perturbations**. (*Under Review*)
- **Chhetri, A.**, Korhonen, J., Gyawali, P., & Bhattarai, B. (2025). **NERO: Explainable Out-of-Distribution Detection with Neuron-Level Relevance in Gastrointestinal Imaging**. *In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2025*

SELECTED ACADEMIC PROJECTS

Improving Adversarial Image Robustness For Preventing Image Manipulation

Oct. 2023 - Apr. 2024

Capstone Project

[Link to Repo](#)

- Developed a **JPEG-resistant image immunization pipeline** to prevent text-guided AI manipulation, specifically for text-guided diffusion models.
- Investigated the impact of text prompts on the **success of adversarial attacks** against **text-guided diffusion models**.

Real-Time Visual Element Generation Using GANs

Dec. 2022 - Apr. 2023

Academic Project

[Paper](#) | [Link to Repo](#)

- Developed a **real-time neural rendering system** for generating realistic grass in video games.
- Experimented with existing architectures, performed result analysis, and designed new architectures that achieved both visual fidelity and low inference time.

TEACHING EXPERIENCE

Fifth Annual Nepal AI School - NAAMII

27th Dec., 2024 - 6th Jan., 2025

Teaching Assistant

Certificate

- Performed live demonstrations of adversarial attacks in image models and LLMs to an audience of 100 students under the supervision Prof. Yash Raj Shrestha (University of Lausanne, Switzerland).
- Assisted in lab sessions on Geometric Deep Learning, Active Learning, and Self-Supervised Learning.

AWARDS AND HONORS

Nepal-US Hackathon - **Category Winner (Innovative)**

March 2026

Nepali Leaders Network

Mentee, Women Leaders Mentorship Programme, *EU in Nepal*

Dec. 2024–October 2025

Mentor: **H.E. Veronique Lorenzo**, EU Ambassador to Nepal, **4% acceptance rate**

Medical OOD Analysis Challenge, **5th Place**, Pixel-level Task

October 2024

MICCAI 2024

Hacking for Humanity 2023, **2nd Runner Up**

February 2023

Sheqal Foundation, Nepal's largest all female hackathon

GritFeat AI Health Hackathon, **1st Position**

January 2023

GritFeat Solutions

ABU-Robocon 2022, **2nd Runner Up**

August 2022

Asia-Pacific Robot Contest

3rd Annual Nepal AI School - **Full Scholarship**

December 2021

Nepal Applied Mathematics and Informatics Institute for Research

Mahatma Gandhi Scholarship

2017-2018

Embassy of India

VOLUNTEER EXPERIENCE

• Organizing Committee

Oct. - Dec., 2024

Fifth Annual Nepal AI School - NAAMII

- **Lead Organizer** for the AI School Hackathon, where I managed logistics, including stakeholder coordination, task division, and event planning both prior to and during the event.
- **Served on the admissions committee** and evaluated around 150 applications under scholarship category.

• AI Trainer, Right to Information

10th Aug. - 14th Aug., 2024

Sancharika Samuha, The Asia Foundation, USAID Nepal

- Led an AI training session in eastern Nepal with **50 journalists from 6 districts**, focusing on enhancing their understanding of AI technology.
- Provided foundational AI knowledge and conducted hands-on training, demonstrating how tools like **ChatGPT, DALL-E, and Claude** can be used by journalists in Nepal to enhance productivity.

• Children in Technology, 2023

26th - 29th Nov., 2023

LOCUS 2023

- Led a team of **13 undergraduate students** to **5 underserved rural communities** in eastern Nepal, engaging with **13 schools** and approximately **2,000 students** on internet safety and cyber law.

AFFILIATIONS

• Event Manager

Dec. 2021 - Feb. 2023

IT Club, Pulchowk Campus

- Organized **5 events**, each attended by approximately **40 students**, covering topics such as low-level system programming and UI/UX design.
- Oversaw **technical setups** and **logistics** for all club events.

• Student Member

Dec. 2019 - Aug. 2022

Robotics Club - Pulchowk Campus

- Worked on multi-view geometry and object localization using stereo vision for a ball-throwing robot in **ABU-Robocon 2022**.
- Organized a **robotics training camp** for 35 primary school students at 2 public schools, aiming to encourage early interest in robotics and STEM.
- Mentored junior members for **ABU-Robocon 2023**.